MIL-T-13458B(MU)
16 April 1968
SUPERSEDING
MIL-T-13458A(MU)
18 October 1963

MILITARY SPECIFICATION

TELESCOPES PANORAMIC: M12A7 SERIES

1. SCOPE

1.1 This specification covers a series of panoramic telescopes, hereinafter referred to as telescopes. These telescopes are essentially identical except for reticle pattern and the zero azimuth setting (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments: General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F7687260	Telescopes, Panoramic: M12A7C, M12A7D, M12A7G, M12A7H, M12A7Q, and M12A7S
F7687263	Telescope, Panoramic: M12A7F
F8228975	Telescope, Panoramic: M12A7K and M12A7R
F10555616	Set-up and Operating Instructions for Panoramic Telescopes M12A7 Series

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PACKAGING DATA SHEETS

3444632	Telescope, Panoramic M12A7K
3444633	Telescope, Panoramic M12A7H
7687260	Telescope, Panoramic M12A7C
7687261	Telescope, Panoramic M12A7D
7687262	Telescope, Panoramic M12A7G
7687263	Telescope, Panoramic M12A7F
8213035	Telescope, Panoramic M12A7Q
8213036	Telescope, Panoramic M12A7R
8213037	Telescope, Panoramic M12A7S

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The telescope shall be manufactured in accordance with drawings applicable to the telescope specified in the contract or order and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-O-13830 and MIL-F-13926:

3.3.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated reticle

3.3.2 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The telescope shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

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3.4.2 Operating temperatures.- The telescope shall meet the requirements of 3.20 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.4.3 Sealing.- The telescope shall show no evidence of leakage when subjected to Procedure I of the rain test specified in MIL-F-13926.

3.4.4 Vibration.- There shall be no evidence of physical failure and the telescope shall meet the following requirements of this specification after being subjected to a vibration of 30 cycles per second at an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds.

3.5 Cleanliness.- The telescope shall meet the cleanliness requirements specified in MIL-O-13830. There shall be no more than two particles of dirt visible on the surfaces of the reticle. The area of each particle shall not be in excess of the area of a size ten dig. There shall be no other dirt within the telescope which interferes with observation or adversely affects image quality.

3.6 Orientation.- The telescope shall meet the applicable requirements of this specification with the horizontal locating surface of the shank horizontal and the line of sight positioned in azimuth as indicated in Figure I for the telescope model specified.

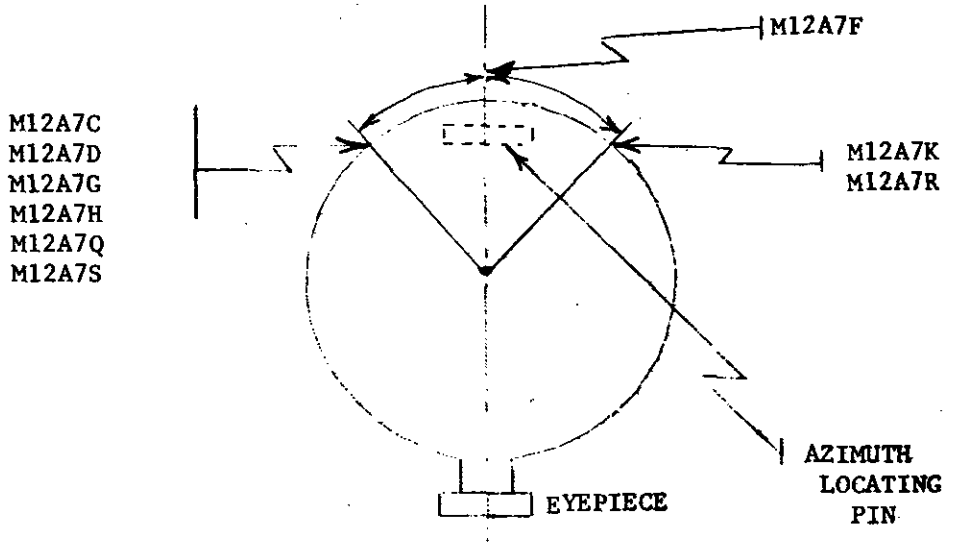


FIGURE I

ZERO AZIMUTH SETTINGS IN RELATION TO AZIMUTH LOCATING PIN

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3.7 Collimation.

3.7.1 Elevation.- With the telescope positioned as specified in 3.6 and the elevation indexes indicating zero, the line of sight shall be horizontal within 0.5 mil.

3.7.2 Azimuth:- With the telescope positioned as specified in 3.6, the azimuth scales shall indicate zero within 0.5 mil.

3.7.3 Open sight.- With the telescope positioned as specified in 3.6, the line of sight of the open sight shall be horizontal within 10 mils and parallel in azimuth to the optical line of sight within 10 mils.

3.8 Image tilt.- The image of a vertical target shall be vertical within 1.0 degree of arc at any azimuth setting.

3.9 Parallelism of reticle and image.- With the horizontal locating surface of the shank horizontal, the center vertical line of the reticle shall be parallel to the image of a vertical target within 30 minutes at any azimuth setting.

3.10 Plumb travel.- With the horizontal locating surface of the shank horizontal and coincidence established with a vertical target at 300 mils elevation, the line of sight shall not deviate more than 1.0 mil from the target through 600 mils travel in depression. The total lateral spread shall not exceed 1.5 mils.

3.11 Backlash.- Backlash shall not exceed 1.0 mil in the elevation worm mechanism; nor exceed 0.5 mil in the azimuth worm mechanism.

3.12 Vertical displacement.- Vertical displacement, due to reversing the azimuth motion by means of the azimuth worm knob, shall not exceed 0.5 mil.

3.13 Level travel.- With the horizontal locating surface of the shank horizontal and coincidence established with a horizontal target, the line of sight shall not deviate more than 1.0 mil from the target through two full revolutions of the head. This requirement shall be met by utilizing the azimuth worm to rotate the head in azimuth.

3.14 Azimuth error.- The error in any azimuth reading shall not exceed 1.5 mils including backlash.

3.15 Scales.

3.15.1 Deflection.- The ratchet on the adjustable micrometer index shall hold the index in coincidence with each graduation of the deflection scale on the worm shoe within 0.2 mil.

3.15.2 Elevation.- With the elevation worm knob indicating zero, the index on the ring shall be in coincidence with the zero line on the head within one half the width of the graduation line.

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3.15.3 Azimuth.- With the telescope positioned as specified in 3.5, the index on the collar (Part No. 7687281) shall be in coincidence with the zero graduation line on the scale within one half the width of a graduation line.

3.16 Eyepiece focus.- Eyepiece focus of the reticle shall be fixed between the limits of -0.75 and -1.0 diopter.

3.17 Parallax.- When viewing a target at 140 ± 10 meters, parallax at the center of the field shall not exceed 0.15 mil.

3.18 Resolution.- The resolution on the optical axis shall be 10 seconds of arc or less. Resolution consists of detectable line structure and proper line count in all four meridians at one focus setting of the observation telescope.

3.19 Reticle.

3.19.1 Accuracy.- The angular values between the zero and maximum graduations in azimuth of the reticle shall be as indicated on the applicable reticle drawing within 2.0 percent.

3.19.2 Illumination.- With the reticle illuminated, the markings shall appear clearly defined when observed in a darkened room.

3.20 Operability.

3.20.1 Knobs.- The running torque required to rotate the adjustment knobs at standard ambient temperature shall not exceed 6 inch pounds for the azimuth worm knob; nor exceed 4 inch pounds for the elevation worm knob. The torque required at the temperatures specified in 3.4.2 shall not exceed the values obtained at standard ambient temperature by more than 50 percent.

3.20.2 Rapid traverse.- The throwout lever shall be manually operable and shall permit manual traverse of the head in azimuth at the temperatures specified in 3.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) panoramic telescopes and three (3) each of every component and subassembly covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Dcouments (IEL) when provided with the technical data package (TDP). Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I and II. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.0 DEFECTIVE		
101. Post vibration	3.4.4	4.6.3
102. Cleanliness	3.5	4.6.3.1
103. Elevation	3.7.1	10555616, Item 1

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104.	Azimuth	3.7.2	10555616,	Item 2
105.	Image tilt	3.8	"	Item 13
106.	Parallelism of image and reticle	3.9	"	Item 3
107.	Plumb travel	3.10	"	Item 11
108.	Backlash	3.11	"	Item 4
109.	Vertical displacement	3.12	"	Item 5
110.	Level travel	3.13	"	Item 6
111.	Azimuth error	3.14	"	Item 6
112.	Reticle illumination	3.19.2	"	Item 15
113.	Deflection scale	3.15.1	"	Item 7
114.	Elevation scale	3.15.2	"	Item 7
115.	Azimuth scale	3.15.3	"	Item 7
116.	Eyepiece focus	3.16	"	Item 9
117.	Parallax	3.17	"	Item 10
118.	Resolution	3.18	"	Item 12
119.	Reticle accuracy	3.19.1	"	Item 8
120.	Rapid traverse (60°-90°F)	3.20.2	"	4.6.3.2.1
121.	Knob torque (60°-90°F)	3.20.1	"	4.6.3.2
MINOR:				
201.	Open sight	3.7.3	"	Item 14

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Environmental sampling.- Sampling shall be selected at random from lots in accordance with inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The control samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and test.

TABLE II

301.	Sealing	3.4.3	4.6.1.1
302.	Storage temperature	3.4.1	4.6.1.2
303.	Operating temperature	3.4.2	4.6.1.2
304.	Knob torque (-40° + 150°F)	3.20.1	4.6.3.2
305.	Rapid traverse (-40 + 150°F)	3.20.2	4.6.3.2.1

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for b contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental test.

4.6.1.1 Sealing.- Perform the rain test as outlined in Procedure I of MIL-F-13926. At the conclusion of the rain test, subject the panoramic telescope to the temperature test procedure described below. When the telescope is thermally stabilized at the high operating temperature specified in 3.4.2, use the shadowing technique to determine that moisture has not gathered on the optic to assure conformance with 3.4.3.

4.6.1.2 Temperature test.- The units under test shall be subjected to Test Procedure I of MIL-F-13926, however, the temperature specified in 3.4.1 and 3.4.2 shall apply.

4.6.2 Performance test.

4.6.2.1 All panoramic telescopes shall be oriented in accordance with 3.6 when positioned as specified on Drawing F10555616. Acceptable inspection procedures are also specified to assure conformance with the detail performance requirements.

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4.6.2.1.1 This specification covers a series of panoramic telescopes having a different zero azimuth setting with respect to the eyepiece. The operating procedure specified on Drawing F10555616 applies to the following models M12A7-C,-D,-G,-H,-Q,-S. When performing acceptance inspection on models M12A7K and R, paragraph 1 of operating instructions shall be changed to read 45 degrees to the right and 45 degrees in a counterclockwise direction. When performing acceptance inspection on model M12A7F, rotation of the azimuth plate is not required.

4.6.2.1.2 In addition to the suggested test procedure specified on Drawing F10555616, the procedure outlined in 4.6.1 and 4.6.3 shall apply.

4.6.3 Vibration.- This test to be performed in accordance with Test Procedure V of MIL-F-13926 except that test duration shall be 5 ± 1/4 minutes. At the completion of the vibration period the instrument shall be examined using visual and tactile type inspection techniques to assure conformance with 3.4.4.

4.6.3.1 Cleanliness.- Cleanliness shall be determined using the Shadowing Technique specified in MIL-O-13830. Cleanliness shall be within the limits specified in 3.5.

4.6.3.2 Torque.- This test shall be performed using a commercial type inch pound torque wrench with adapters to accommodate the azimuth and elevation knobs. Readings with the torque wrench shall be taken in both directions and while in motion. The running torque required to rotate in the azimuth and elevation knobs at standard ambient temperature and at the temperatures indicated in 3.4.2 shall not exceed the values as specified in 3.20.1.

4.6.3.2.1 Rapid traverse.- By actuating the throwout lever the head shall be free to be turned in azimuth in both directions with no binding. This test shall be performed at standard ambient temperature and at the temperatures specified in 3.4.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with the applicable packaging data sheet listed in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The panoramic telescopes of the M12A7 series are primarily intended to lay the weapon in azimuth for indirect fire. The model number identifies the reticle pattern and orientation of the eyepiece relative to the zero azimuth setting.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.

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(c) Applicable packaging data sheet number.
(d) Applicable stock number.

International interest.- Certain provisions (Section 3) of this specification are the subject of international standardization agreements (ABCA-Army-STD-56, 57 and 98). When amendment, revision, or cancellation of this specification is proposed, the departmental custodian will inform its Departmental Standardization Office so that appropriate action may be taken respecting the international agreement concerned.

Custodian:

Army - MU

Preparing activity:

Army - MU

International interest (see Section 6).

Project No. 1240-A081

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Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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